

## Risk Assessment and Mitigation

| Criteria   | Risk Level    |
|------------|---------------|
| 1 = Low    | 1 = Very Low  |
| 2 = Medium | 2 = Low       |
| 3 = High   | 3, 4 = Medium |
|            | 6 = High      |
|            | 9 = Critical  |

| Risk                                                                          | Risk Category | Probability | Impact | Risk Level | Mitigation                                                                                                                 |
|-------------------------------------------------------------------------------|---------------|-------------|--------|------------|----------------------------------------------------------------------------------------------------------------------------|
| AI credits are too expensive to build and test the AI scenario                | Financial     | 3           | 3      | 9          | Check we can get free access to LLMs for educational purposes or if BFH has alternative solutions.                         |
| Not enough time for documentation                                             | Operational   | 2           | 3      | 6          | Add a dedicated documentation issue with an according weight to each epic.                                                 |
| Not enough time for testing (drills, scenarios and environments)              | Operational   | 3           | 2      | 6          | Add a dedicated testing issue with an according weight to each epic.                                                       |
| Simulating or building network environments needs more time than planned      | Operational   | 3           | 2      | 6          | Add dedicated issues with according weight to each scenario epic.                                                          |
| User does not understand how to operate NSAK                                  | Operational   | 3           | 2      | 6          | Provide a Web GUI and/or user guides                                                                                       |
| Scope is too large for bachelor thesis                                        | Strategic     | 2           | 3      | 6          | We have to plan and evaluate the goals carefully. The use of "User Story Maps" also helps to mitigate this risk.           |
| Scenario or drill too complex for automation                                  | Strategic     | 2           | 3      | 6          | Create "User Story Maps" for epics, so that we have a good understanding of the scenarios and their drills.                |
| The AI bubble bursts and we need to switch AI provider                        | External      | 2           | 2      | 4          | Use vendor agnostic tools and protocols, so that we can switch provider easily.                                            |
| Complex network environments need more hardware than available                | Financial     | 2           | 2      | 4          | We can simulate network environments instead of real hardware.                                                             |
| Scenarios and drills need more time than planned                              | Operational   | 2           | 2      | 4          | Use "User Story Maps" to get a good idea of the required steps to build the scenario, it's drills and estimate generously. |
| Not enough time for media artifacts (poster, presentations, film, book, etc.) | Operational   | 2           | 2      | 4          | Include the creation of the media artifacts as milestones/epics in the project planning.                                   |
| Misaligned communication leads to wrong expectations                          | Strategic     | 2           | 2      | 4          | Multiple channels and Bi-Weekly Meetings should keep the communication aligned.                                            |
| One of the students gets injured or sick                                      | External      | 1           | 3      | 3          | We accept this risk.                                                                                                       |
| Regressions: tested scenarios stop working because a drill was changed        | Operational   | 3           | 1      | 3          | Add automated test cases for detecting regressions early.                                                                  |
| The NSAK framework/device itself is vulnerable                                | Operational   | 3           | 1      | 3          | Clearly state that the NSAK framework/device is still a POC and not ready for production use.                              |
| The NSAK hardware devices stop working                                        | Financial     | 1           | 2      | 2          | The fact that we have two devices already reduces the impact, further we could also use other HW.                          |
| Drills not compatible between scenarios                                       | Operational   | 2           | 1      | 2          | The implementation of the configuration management should provide a way that small deviations can be branched.             |
| The AI might go rogue in the AI scenario                                      | Operational   | 2           | 1      | 2          | Besides the fact that we have physical access to the device we add a killswitch to the AI scenario.                        |
| An existing framework already does everything we have planned                 | Strategic     | 2           | 1      | 2          | Compare well known frameworks with NSAK and highlight our USP (HW and AI).                                                 |
| NSAK gets abused by hackers                                                   | External      | 1           | 1      | 1          | Clearly state that the NSAK framework/device is built only for educational and white hate purposes.                        |
| Lack of resources for attack scenarios                                        | Operational   | 1           | 1      | 1          | Get familiar and use the MITRE ATT&CK framework as reference and source.                                                   |
| Lack of resources / consensus for defense scenarios                           | Operational   | 1           | 1      | 1          | Get familiar and use well known sources such as NIST as source.                                                            |